

Mathematics for Data Science

Session 1 - Course logistics

Eva FEILLET¹

¹LISN

Université Paris-Saclay, CNRS,
91190 Gif-sur-Yvette, France

M1, Fall 2026

Table of Contents

- 1 Motivation
- 2 Objectives of the course
- 3 Organization
- 4 Resources

Table of Contents

1 Motivation

2 Objectives of the course

3 Organization

4 Resources

Data Science relies on Math

Need to represent the data:

- Tabular data: a data point can be represented as a *vector*, a data set as a *matrix*
- Image data: an RGB image can be represented as a 3D tensor, or as a vector. . .
- Text data: how to represent words? and sentences?
 - bag of words and TF-IDF representations
 - word embeddings (word2vec, fastText. . .)
 - sentence embeddings (BERT, GPT. . .)
- Use “encoders” i.e. *functions* that map data points into a “representation space”

Data Science relies on Math

Need to manipulate the data:

- Compare data points e.g. using the cosine similarity, an L2 distance, Mahalanobis distance. . .
- Aggregate data points e.g. computing a mean vector, a covariance matrix, a Gram matrix. . .
- Transforming data e.g. pre-processing image data (transformation on color channels, crop, distortion)
- Performing dimensionality reduction: as a pre-processing step before learning, or for visualization e.g. PCA, but also t-SNE, UMAP. . .

Data Science relies on Math

Need to learn from the data:

- Analytic solutions e.g. linear regression
- Distance-based classifiers using mean embeddings and covariance matrices
- Backpropagation in neural networks: chain rule, optimization
- Neural networks: parameters/weights = matrices.
 - How to initialize weights ? e.g. Xavier initialization, Kaiming initialization. . .
 - How to design the architecture of a neural network ? e.g. number of layers, number of neurons per layer. . .
 - How to design the structure of some particular weight matrices ? e.g. butterfly matrices, Toeplitz matrices. . .
 - How to compress weights ? e.g. low-rank approximation of weight matrices, pruning. . .
 - How to determine which weights are important ? e.g. Fisher information matrix, Hessian matrix. . .

Table of Contents

1 Motivation

2 Objectives of the course

3 Organization

4 Resources

Objectives of the course

Refresher course: review the main concepts useful in data science
Lectures about linear algebra (x5) and multivariable calculus (x1)

Objectives of the course

Refresher course: review the main concepts useful in data science
Lectures about linear algebra (x5) and multivariable calculus (x1)

Covered topics

- (Lecture 1) Vector spaces and manipulation of vectors
- (Lecture 2) Matrices and linear systems
- (Lecture 3) Determinants
- (Lecture 4) Eigenvalues and eigenvectors, diagonalisation
- (Lecture 5) Norms, scalar product, orthogonality
- (Lecture 6) Taylor formula, partial derivatives, differentiability

Objectives of the course

Yes, ChatGPT can help you with linear algebra and calculus...

**WHEN THE ONLY
PROGRAMMING LANGUAGE
YOU KNOW IS CHATGPT**



...but it is important to understand the concepts behind the calculations.

Table of Contents

- 1 Motivation
- 2 Objectives of the course
- 3 Organization**
- 4 Resources

Course organization

6 × (1h30 of lecture + 1h30 of practical session)

Every FRIDAY / 1.30pm to 5pm from 11.th of Sept to 23rd of Oct.

Evaluation :

- written exam (2h): **23rd of October**
- closed books

Table of Contents

- 1 Motivation
- 2 Objectives of the course
- 3 Organization
- 4 Resources**

Course homepage

Course homepage on e-campus : Register here TODO QRCode

Alternative course homepage : <https://github.com/EvaJF/math4DS/>

Useful links and resources to complement the lectures

- Lectures "Introduction to Probability, Statistics, and Machine Learning" by Samuel S. Watson from Brown University (lectures 2, 3, 4), <https://data1010.github.io/class/>
- Recap "Mathematics for Machine Learning" by Garrett Thomas from University of California, Berkeley, <https://gwthomas.github.io/docs/math4ml.pdf>
- Book "Mathematics for Machine Learning" by M. P. Deisenroth, A. A. Faisal, C. S. Ong, published by Cambridge University Press, <https://mml-book.github.io/book/mml-book.pdf>

Contact

Mail subject should begin with [***math4DS***]
Ask questions in class :-)